Econ 393BC: Econometrics - Fall 2023

Colby College Department of Economics

Instructor: Dan LaFave **Email:** drlafave@colby.edu

Contacting me: Please use the following guide to get your questions answered:

1. I have a question that is useful for other students as well:
    a. Please post it to the course Question Forum on Moodle. I'll answer it there.
2. I have a question about course material that is specific to me or a question, comment, or concern I'd like to only send to Prof. LaFave:
    a. Please email me and I'll get back to you as soon as possible

Classrooms: Section B Tu/Th 9:30am – Diamond 341
Section C Tu/Th 11:00am – Diamond 342

Office Hours: W 2:00-4:00pm; Th 2:00-4:00pm; anytime my door is open; and by appointment
Please visit me in Diamond 351

Class Materials: Posted on weekly basis to Moodle

TAs: Grace Hillis and Fiona Nash - Help sessions: Th 7-10pm Diamond 122

Econometrics in a post-pandemic world

I'm committed to ensuring that you learn everything you hoped to learn from this class. I will make whatever accommodations I can to help you finish your work, do well on your projects, and learn and understand the material. I understand we all face difficult challenges, especially under (post)-pandemic conditions, and I am here to support you.

I will not judge you or think less of you if you tell me you're having trouble.

You *never* owe me personal information about your health (mental or physical). You are *always* welcome to talk to me about things that you're going through. I will find somebody who can help if I cannot.

If you need extra help or time, or if you feel behind and that you're not understanding everything, **do not suffer in silence!** Please talk to me. I **promise** I will work with you.

I want you to learn how to do and understand econometrics during this class, but I primarily want you to stay healthy, balanced, and grounded during this semester.

Course Description

This course explores the measurement and evaluation of economic models and causal effects, and puts what you've learned in other core economic courses to action. Learning econometrics is a hands-on endeavor, and requires the use of economic theory, statistical analysis, and computer programming. The emphasis of the course is on intuitive understanding and the application of econometric techniques to real world problems encountered in public policy, government, and business.

We focus on causality throughout. We begin by reviewing the linear regression model with a number of simplifying conditions that lead to very nice statistical properties – unbiasedness and efficiency. We then proceed to investigate econometric analyses when the conditions do not hold, and learn techniques developed to handle various types of real-world issues.

Econometrics requires developing and implementing a wide skill set: mathematics, statistics, economic theory, programming, written and verbal communication, and (perhaps most of all) intuitive sense. Be prepared for an engaging and rigorous semester.

Pre-requisites
I expect that you have completed and are comfortable with Intermediate Microeconomics EC223 and Research Methods and Statistics EC293. If this is not the case, please speak to me directly. The course will draw heavily on the foundation developed in EC293. While we will review some of the key concepts at the beginning of the semester, the pace of the course dictates only a small time to review – if you feel particularly rusty, please schedule a time to meet.

Goals of the Course

- Develop your ability to critically evaluate economic research and commentary.
- Learn practical skills for data analysis.
- Improve your ability to produce, interpret, and communicate your own research.
- Understand and apply the intuition behind econometric analysis and statistical inference to a number of real-world problems.
- Participate in the practice of economics as it is applied throughout professional settings.

Expectations

Understanding what I expect of you and what you can expect from me is essential for a successful course - not only for doing well in terms of your grade, but also for having an engaging and dynamic class experience.

What I expect from you:

- I expect you will uphold the Colby Affirmation and demonstrate academic honesty in all parts of the course - not only on exams. Students found to be in violation will receive a zero on the assignment and be referred to the appropriate Dean. Colby takes these issues seriously.
- Maintain an active engagement in the course. Pay attention and respond when called upon. Be engaged outside of class as well; follow along with the course content, prepare for exams and complete assignments by their due dates.
- Come to class prepared: ****It is <u>essential</u> that you come having looked at the posted summary material.**** The course moves at a pace that reflects this expectation.
- Help guarantee that we create an inclusive environment where all students are well-served by this course, and one that recognizes the strength and benefit of diverse perspectives.
- Communicate any anticipated absences to me in advance
- Actively participate in the learning process. Help each other to establish a collaborative learning environment. Ask questions when you don't understand something. Come to TA and/or office hours if you're confused – use our available resources to set yourself up for success.

What you can expect from me:

- I will come to class prepared, use class time effectively, and foster a learning environment that is engaging, inclusive and respectful. I will gladly honor your request to address you by preferred pronoun and/or an alternate name than what is listed on the official class roster– please advise me of this early in the semester.
- I will present econometrics so that it is relevant to your lives, not only a set of abstract concepts and formulas. I will work with you to build knowledge and develop a new lens through which to view the world.
- Assess your learning fairly and provide constructive feedback on all quizzes and exams.
- Be available and approachable. I will be available for office hours and work with those who have time conflicts to find meeting times that work for the both of us. I will do my best to respond to all emails within 12 hours.

Health related policies

- Please do not come to class if you feel unwell.
- Should you need to miss class due to health related concerns, I will work with you to make sure you maintain progress in the course.
- There is a non-trivial possibility we will have to have a remote class or two due to my parenting responsibilities and school/daycare closures. I will let you know as far in advance as possible, but it may be short notice.
- I will not let anything derail your econometric experience

Grading

The majority of work in the course will be staying on top of the material. This is no small task - we will cover *a lot* of material. Developing your econometric intuition is the best way to succeed. To facilitate this, your active participation in class discussions is expected, and you are encouraged to speak up with questions. Out of class assignments will complement the course material.

- **Problem Sets:** Each module is capped with a set of exercises focusing on implementing and interpreting the weekly topics. These are due via Moodle at the specified date and time. We all have things that pop up and derail even our best-laid plans, so I'll drop your lowest problem set grade over the course of the term, but they **will <u>not</u> be accepted late**. Solutions will be posted on the course webpage.

  You're more than welcome to work with others on these exercises - explaining an idea, concept, method or result to a peer is one of the best ways to reinforce your own understanding. However, to be sure that you do understand the work, *you must write up your answers in your own words and submit your own work*. This applies to AI-based assistance as well. Please let me know on your problem sets whom you collaborated with and what AI tools you used, if any.

  Problem sets are changed year-to-year. To avoid confusion and disappointment, prior year answer keys may not be used under any circumstances. If a violation occurs, a grade of zero will be assigned to **<u>all</u>** of your problem sets for the semester and an Academic Dishonesty report filed with the Dean.

Building on your work in EC293, a portion of each problem set will require you to use Stata – a general-purpose statistical package used across the social sciences, government agencies, consulting firms, and financial corporations. There are extensive web-based resources to assist you with using the program, and we will go over some of the basics in class. You do not need to purchase STATA – it is available in campus computer labs and online via the Colby Keyserver. It is fine to use R/Python/some alternative if you wish.

- **Midterm Exam:** We will have one midterm exam on **October 19th**.

- **Research Project:** The capstone for the course will be a research project completed in assigned groups of **four** on a topic of your choice. This will be a practical application of econometrics to look at a research question your group finds interesting. This is a very rewarding experience, but will also be a significant time commitment. To spread the work out over the course of the semester, we'll use the following timeline:

	Thursday	**September 7th**	Day #1
1.	**Friday**	**Oct 6st**	Submit abstract of your topic
2.	**Friday**	**Oct 27th**	Data summary
3.	**Friday**	**Nov 17th**	Initial regression report– detailed data analysis, substantial progress on each section
4.	**Thursday**	**Dec 7th**	Research poster for final day of class
5.	**Monday**	**Dec 18th**	Final results report

We will discuss this project at length throughout the semester. To get you started, a handout is posted on Moodle to aid in topic selection.

- Your final grade will be determined according to the following weights:
  - Participation (includes looking at pre-class material): 15%
  - Problem Sets/Weekly exercises: 15%
  - Midterm Exam: 20%
  - Final Exam: 25%
  - Research Project: 25%
    - Abstract: 3% Data Summary: 4% Initial Report: 5%
    - Final poster: 6% Final report: 7%

- *A Note on Re-grade Requests:* You may resubmit any exam or problem set with an additional sheet attached explaining in a few words why you deserve a better grade. These are due one week after the graded work is returned. If you choose to petition for a change to your grade, I will re-grade the **entire** work – not only the part in question.

Texts and Readings

The most important material to look at are the lecture notes I will post on Moodle. **I expect that you will look at the relevant posted material before coming to class.**

Primary textbook: Jeffrey Wooldridge. *Introductory Econometrics: A Modern Approach*

Supplemental: C.F. Baum, *An Introduction to Modern Econometrics Using Stata.*
Angrist and S. Pischke, *Mostly Harmless Econometrics.*

You can borrow both of these texts from my office and they are also available in Miller – the first is a useful guide for getting started with STATA, and the later does an excellent job of presenting regression analysis in an intuitive way.

Supplementary Readings:

Any major newspaper. Econometrics is all around you – nearly every day there will be an article in Section A of the newspaper that relies on some sort of statistical analysis we will encounter in class. I encourage you to take advantage of Colby's licenses for both the New York Times and Wall St. Journal – the College provides free access to both.

Additional course material – lecture notes, handouts, etc. – will be posted on the course Moodle page.

Course Outline *(tentative and subject to change)*

1. Introduction and Statistics and Probability Review

Ch. 1 Nature of Econometrics
Appendix A, B, C
Ch. 19 Econometric Research

Key Concepts: *Conditional probability; Regression and relationships between variables; Econometric Methodology; Causality vs. Correlations*

2. Classical Linear Regression Model and Ordinary Least Squares

Ch. 2 The Regression Model
Appendix A.5

Key Concepts: *Intro/Review of Stata; OLS; Derivation of OLS; R^2*

3. Econometric Analysis of Cross-Sectional Data

Ch. 3 Multiple Regression Estimation
Ch. 4 Hypothesis testing
Appendix C.5-C.6
Ch. 7 Qualitative Data

Key Concepts: *t-test; Confidence intervals; Hypothesis testing; F-tests; Adjusted R^2; Elasticities; log-linear; semi-log; Program evaluation; Treatment effects; Dummy variables; Linear probability models*

4. Midterm Exam

5. Real Life Regression Issues - Nature and Causes of Bias

Ch. 6 Additional Issues in OLS
Ch. 8 Heteroskedasticity
Ch. 9 Specification & Data Problems

Key Concepts: *Interaction effects; Bias; Consistency and efficiency; Multicollinearity; Specification errors; Heteroskedasticity; Robust standard errors; Outliers; Least Absolute Deviation*

6. Nonlinear Models with Limited Dependent Variables

Ch. 17 Limited Dependent Variables

Key Concepts: *Logistic regression; Probit regression; Maximum likelihood estimation; marginal effects*

7. Techniques for Panel Data

Ch. 13 Basic Panel Methods
Ch. 14 Advanced Panel Methods

Key Concepts: *Pooled cross-sections; Unobserved heterogeneity; Fixed-effects; Within variation; First-difference models; Program evaluation; treatment effects; Difference-in-Difference; Clustering*

8. Machine Learning tools for Causal Analysis

Lecture notes

Key Concepts: *Machine learning, artificial intelligence, variable selection, LASSO*

9. Analysis of Time-Series Data

Ch. 10 Basic Time-Series Methods
Ch. 11 Further Issues with Time-Series
Ch. 12 Serial Correlation

Key Concepts: *Time trends; First-difference models; Serial correlation; Spurious relationships*

Final Exam: TBD



EC393: Econometrics Info Sheet - Fall 2023

Name _____ **Year**_____

Major/Minor_____

Hometown or fun fact about yourself (helps me remember your name)_____

_____

I have taken: ☐ **EC293** ☐ **MA381/SC482** ☐ **Something else**

During what semester/year? _____

And: ☐ **Loved It** ☐ **Really Loved It** ☐ **Other**

Describe your comfort with STATA .do files:

☐ **Never Learned** ☐ **Knew them at some point** ☐ **Comfortable**

When was the last time you ran a regression? _____

Why are you taking this class? (Honesty counts)_____

_____

What questions interest you in economics? _____

_____

_____

_____

On what topics would you like to see examples and applications of regressions? _____

_____

When would you prefer your weekly exercises are due? We start them in class on Thursdays.

☐ **Friday @5pm** ☐ **Saturday @5pm**